

Md Tanzeel Adam Khan

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Education

Indian Institute of Technology Patna <i>Master of Technology in Artificial Intelligence (CGPA: 8.5/10)</i>	2024 – 2026 Patna, India
Haldia Institute of Technology <i>Bachelor of Technology in CSE – AI & ML (CGPA: 8.9/10)</i>	2020 – 2024 Haldia, India

Publication

When Should Language Models Search? A Topology-Aware Analysis | *Under Review, TMLR*

- First Author | Proposed A* search over LLM reasoning traces with formally proven admissible heuristic ($h(s) \leq h^*(s)$). Evaluated across **8 benchmarks** $\times$ **3 scales (0.5B–14B)** $\times$ **4 methods** = 96 configurations, 500 instances each.
- Designed a zero-LLM-call Topology Router using 14 structural features; achieves **40% oracle gap recovery** and **95% routing accuracy** on CodeContests. Branching Coefficient correlates at $R^2 = 0.94$.

Research Experience

Indian Institute of Technology Patna <i>M.Tech Thesis Advisor: Prof. Asif Ekbal LLM Reasoning & Search</i>	Jul 2024 – Jun 2026 Patna, India
- Implemented A* search with real branching ($K=3$ separate LLM calls at varied temperatures), self-consistency voting over multiple goal nodes, entity-coverage heuristic, and degenerate-output detection. Ran 48,000+ paired evaluations across 8 benchmarks with Wilson confidence intervals and McNemar significance tests. - Engineered topology-aware routing pipeline: feature extraction (13 structural signals), multi-classifier optimization (XGBoost/GradientBoosting, 5-fold CV, 100 random seeds), and complementarity decomposition analysis. Discovered Fluency-Correctness Asymmetry: 38% of fluent traces are incorrect — explaining why surface-confidence routers fail.	

Professional Experience

Mercedes-Benz Research & Development India (MBRDI) <i>AI Engineer Intern AMG System Software Validation Team</i>	Nov 2025 – Aug 2026 Bengaluru, India
iFAST (Intelligent Failure Analysis through Signal Tracing): Architected an autonomous AI agent (Claude/AWS Bedrock + 14 MCP tools over JSON-RPC 2.0) performing probabilistic root-cause analysis using Bayesian evidence fusion from 8 independent sources with calibrated Log-Likelihood Ratios. Achieved 54.8% time savings and 97% accuracy across 67K+ signals and 1,128 modules. Adopted by 2+ engineering teams beyond initial scope. - Built a novel source-code dependency graph engine extracting 1.6M+ inter-module relationships from C source via regex parsing across 3 repositories. Signal-level verification eliminates 75% of false positives. Replaced manual export that was 35% incomplete with a live git-based pipeline. RAG Chatbot + KPI Dashboard: Deployed hybrid BM25+ANN retrieval with cross-encoder reranking and LangGraph orchestration over Jira SWF, Jira MCP, and Confluence. 77% answer rate, 5.5$\times$ latency reduction (22s $\rightarrow$ 4s) via semantic caching and async parallelism. ASPICE SPOT Check (SWE.6): Architected A2A system using JSON-RPC 2.0 for automated compliance assessment across 60+ evaluation dimensions with evidence-grounded scoring, hierarchical aggregation, and multimodal ingestion (7 formats).	

Moshi Moshi <i>AI Engineer Intern</i>	Jun 2025 – Aug 2025 Bengaluru, India
- Built an automated lead generation pipeline using Selenium web scraping + open-source LLMs for prospect enrichment and personalized cold email generation via SMTP. Automated 40–60% of the sales workflow end-to-end. - Developed Inspira — a GenAI tool for designers to generate creative website concepts and content for branding clients. Reduced design ideation turnaround by 35% and development handoff time by 25%.	

Technical Skills

ML/DL: PyTorch, HuggingFace Transformers, LoRA/QLoRA/PEFT, XGBoost, scikit-learn, Bayesian Inference
LLM Systems: LangGraph, LangChain, RAG, MCP Protocol, A2A/ReAct Agents, Tool-Use, vLLM, Ollama
Retrieval: BM25, ANN, Cross-Encoders, Reciprocal Rank Fusion, ChromaDB, LanceDB, Sentence-Transformers
Cloud & Infra: AWS (Bedrock), Docker, Git, CI/CD, FastAPI, Streamlit, PySide6/Qt
Engineering: Python, asyncio, SQL, NumPy, Pandas, SciPy, NetworkX, Regex Engines, ProcessPool/ThreadPool

Additional

Teaching: Teaching Assistant – Deep Learning & AI courses, IIT Patna
Certification: Deep Learning Specialization – DeepLearning.AI (Andrew Ng)
Coursework: Deep Learning, Natural Language Processing, Information Retrieval, Probabilistic Graphical Models